

## Education

Universidad de Valparaíso

Valparaíso, Chile

Physics graduate.

Finalization: 20/12/2024 [Thesis: 7.0]

Mention in scientific computing: Unix, numerical methods, database, IA,  
Paralleling programming, Optimization with GPU, deep learning (CNN y PINN).

## Experience

Universidad de Valparaíso

Valparaíso, Chile

Jul 2024 – Dec 2024

Thesis.

- My thesis project has the title: "Simulation and study a gravitation collapse of the no massive scalar field with a Quantum correction".
- I developed a code that simulates a dynamic of the scalar field with quantum correction and how this interacted with the tensor metrics.
- The code is in C and optimization with CUDA, improving the execution time significantly as a function that the mode normal increases, for example, for the case the 20 modes l and k, were originally scheduled for 6 hours and 44 minutes; with my optimized code it was 38 minutes, with a 90.5% improvement. I also excelled in proposing numerical techniques that improved the simulation's stability.
- Obtaining a scholarship from the Center for Theoretical Physics of Valparaíso (CeFiTeV), which was intended to support the thesis project.

Universidad de Valparaíso

Valparaíso, Chile

Class assistant.

Apr 2022– Jul 2022 & Apr 2023 – Jul 2024

- I worked as a teaching assistant for courses including the "Differential equations" (two times), "Thermodynamics" and "Intermediate mechanics".
- I supported in the teaching of physics' students, solved problems and discussing the theory.
- I taught reinforcement and review classes.

## Projects

- Optimization in CUDA of the N-Body simulation:
  - This code was written in C, permitted study the dynamics of many bodies, it was testing with a  $N=100$  and  $N=1000$  using a particle-mesh method.
  - Solved Poisson's equation via Jacobi iterations; error threshold:  $1e-6$ .
  - Accelerated the computation with CUDA, improving performance at scale.
- Data science project about Air pollution in South Korea:
  - Using two real datasets from Korea South, one is data weather and the other data pollution. The purpose was built models for de prediction the 2.5 particulate matter.
  - Realizing ETL with Pandas, merging the datasets, calculated the distance between city and station, was able to identify three cities for the analyzing, each one underwent an EDA (pyplot) showing a pm25 concentration of each city and the other variables correlations.
  - Built Random-Forest model and SVM model for each city, train with cross-validation and according to the metrics the better score was 63%
- Waves simulation's software in python (TKinter).
  - This project aimed to help in the teaching of wave physics to high school and university students
  - The software was shared with physics students for educational use.

### **Rodolfo Godoy Arteaga**

- The program contains sections of “Standing wave”, “Interference”, “Coupled System”, “Spring” and “Types of waves”.

### **Skill & Interest**

#### **Technical:**

- Programming: Python (intermediate), C (intermediate), SQL (intermediate) y CUDA (intermediate).
- Experience using SSH for connect remote servers and run simulations in Linux environment.
- Libraries and APIs: Pandas, matplotlib, TF, scikit-learn (CNN, RNN, PINN, random forest, SVM y Knn), numpy y TKinter.
- Parallelization: CUDA, MPI, OpenMP
- Other tools: Excel, Power BI (basic) y UNIX (ssh).

**Languages:** Spanish (Native) / English (Basic- intermediate): Reading: I can read and understand scientist’s articles and instructions technical (intermediate) y Listening/Speaking (Basic)

**Interest:** Interested in delving deeper into computational physics, machine learning, and advanced simulation methods. I remain focused on physics and hope to continue my studies in the future. I'm constantly exploring new tools like Flask, Qiskit, EinsteinPy, and others.